The University of Texas at Austin

Project Management and Construction Services

Design and Construction Standards

Technical Specification

SECTION 23 36 00

Air Terminal Units

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The issuance and revision history of this Section is tabulated below. Please destroy any previous copy in your possession.

Rev Date	Pages	Remarks	Documents Referenced
11/15/2017	All	New specification section created.	23.36.00 Air Terminal Units.docx
12/21/2017	All	Published new specification.	

SECTION 23 36 00 – AIR TERMINAL UNITS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General Conditions and Division 01 Specification Sections, apply to this Section.

B. Specifications throughout all Divisions of the Project Manual are directly applicable to this Section, and this Section is directly applicable to them.

1.2 SUMMARY

A. Perform all Work required to provide and install the following products as indicated by the Contract Documents with supplementary items necessary for proper installation.

1. Single duct variable or constant volume terminal units.
2. Dual duct variable or constant volume terminal units.
3. Fan powered constant volume terminal units.
4. Integral heating coils.
5. Integral controls.
6. Integral sound attenuator.

1.3 REFERENCE STANDARDS

A. The latest published edition of a reference shall be applicable to this Project unless identified by a specific edition date.

B. All reference amendments adopted prior to the effective date of this Contract shall be applicable to this Project.

C. All materials, installation and workmanship shall comply with the applicable requirements and standards addressed within the following references:

1. NFPA 90A - Installation of Air Conditioning and Ventilation Systems.
2. UL 181 - Factory-Made Air Ducts and Connectors.
3. AHRI Standard 410 - Forced-Circulation Air-Cooling and Air-Heating Coils
4. AHRI Standard 880 – Performance Rating of Air Terminals.
5. AHRI Standard 885 – Procedures for Estimating Occupied Space Sound Levels in the Application of Air Terminals and Air Outlets.
6. ANSI/ASHRAE Standard 130 – Methods of Testing for Rating Ducted Air Terminal Units.
7. ASHRAE 90.1 – Energy Standard for Buildings Except Low-Rise Residential Buildings

1.4 SUBMITTALS

A. Product Data:

1. Shop Drawings of product data indicating configuration, general assembly, access space required for service, and materials used in fabrication.

2. Electronic or Printed Catalog performance ratings that indicate nominal inlet size, CFM, applicable static pressure at the inlet or discharge of terminal unit, and noise criteria with sound octave band and sound decibel test in accordance with ARI 880, for the insulation lining selected.

3. Leakage curves indicating inlet static pressure and actual tested leakage rates shall be submitted for all non-standard or custom-built terminal units.

4. Unit manufacturer shall test and certify that each terminal unit used on this Project has been tested as specified.

B. Record Documents:

1. Submit under provision of Division 01.

C. Operation and Maintenance Data:

1. Operating instructions and maintenance manuals indicating maintenance and repair data, parts lists.

D. Sample Units:

1. One sample, 8 inch size, production run unit of each type shall be submitted for examination and approval by the Engineer, Owner, and TAB Firm.

2. This sample unit shall be submitted in addition to the required written submittal, well in advance of any requirement for installation of units, but absolutely no later than 60 calendar days after the Notice to Proceed with Construction.

3. Contractor shall allow a minimum of three (3) weeks for testing of the sample unit from the time shipped to the TAB Firm. The TAB Firm will test single duct terminal for casing leakage, damper leakage, and the specification requirements. The fan powered terminal will be tested for damper leakage, airflow tracking from minimum primary to maximum primary, discharge pressures from 0.25 inches w.c. to 0.6 inches w.c. with three different airflows (maximum, midpoint, and minimum). The fan powered terminal must maintain its downstream airflow ±5 percent. The fan powered terminal will be observed to maintain the Specification requirement. This period shall restart if the sample unit is rejected and another unit is resubmitted.

4. If rejected for any reason, Contractor shall expedite the documented corrections and shall resubmit a sample unit as soon as possible.

5. Any delay in submittal of the unit for approval shall not be grounds for Contractor's claim of delay. If approved, the unit shall remain in the possession of the Owner at the Project Site for comparison with units as shipped to the Project.

6. Unit(s) shall be installed in the Project, at an accessible, marked location.

1.5 SHIPMENT TESTING PRIOR TO INSTALLATION

A. Shipment Testing: At the Owner's discretion, a minimum of ten (10) percent of each size single duct terminal unit (but no less than one unit of each size on the Project) will be tested at the Project Site for casing leakage and damper leakage. Fan powered terminals units will be tested for damper leakage and for conformance to this Specification. Contractor shall allow sufficient time during construction for the TAB Firm to perform all testing as may be required.

B. Unit Non-Performance:

1. If results of the shipment testing show that any of the units do not perform as specified, then an additional ten (10) percent of each size unit (but no less than one unit of a size, unless 100 percent of the size has been tested) shall be tested.

2. If this testing, in the Owner's opinion, shows that ten (10) percent or more of the units tested do not perform as specified, then 100 percent of all unit sizes shall be tested for conformance with these Specifications.

3. The results of that testing shall be reviewed carefully between the Contractor, manufacturer, Owner, and Engineer. A method of repair or replacement of units will be negotiated. The Owner, however, shall maintain the right of final approval of any proposed solution.

C. Should for any reason, the testing as described in this Section prove that any of the units do not perform as specified, Contractor shall be responsible for all subsequent labor, travel, travel expenses and incidental expenses, penalties, or other costs attendant to any additional testing as described in this Section, or as required to prove that the units perform as specified. This shall include, but not be limited to, the labor, travel and reasonable incidental expenses of not only the Contractor and TAB Firm, but also those incurred by the Owner as may be specifically required for this purpose.

1.6 WARRANTY

A. Provide one year manufacturer's warranty under provisions of Division 01.

PART 2 - PRODUCTS

2.1 MANUFACTURER:

A. Dual Duct Terminal Units: All dual duct terminal units shall be as manufactured by Titus (Model MDV-3100-UT or MDC-3100-UT), Metal*Aire (Series TH500-ECO or Series 400DDUT), or Nailor Industries (3000-UT or 3200-UT), or Price Industries. Note that the model and series numbers listed may differ slightly from catalogue information.

B. Fan Powered Terminal Units: All fan powered terminal units shall be as manufactured by Titus (Model DTFS (Series) or DTFP (Parallel)), MetalAire (Series FCI (Constant Volume) or FVI (Variable Volume)), Price (Series FDCA (Series) or FDV (Parallel))), or Nailor 35 series. Note that the model and series numbers listed may differ slightly from catalogue information.

C. Single Duct Terminal Units: Single duct boxes shall be the single duct version of the dual duct versions described above.

D. No other Manufacturers or models are acceptable. Even though specific Manufacturers may be named herein, the material supplied by any approved Manufacturer shall meet all the provisions of this specification without exception.

2.2 VARIABLE OR CONSTANT VOLUME SINGLE DUCT TERMINAL UNITS:

A. The Contractor shall provide pressure independent single duct variable air volume control assemblies as scheduled with integral attenuator of the sizes, capacities, and configurations shown on the drawings.

B. CASING CONSTRUCTION:

1. The units shall be constructed of a minimum of 22-gauge galvanized steel and internally lined with a minimum of 1 inch thick, three pound per cubic foot density insulation.

a. The insulation shall be foil faced with the edges and seams sealed or "captured", encapsulating all fibers of the insulation.
b. Insulation shall meet NFPA-90A requirements for flame spread and smoke generation and UL-181 requirements for anti-erosion, corrosion and fungus properties.
c. The insulation shall be neatly installed with no rough edges to interrupt the smooth flow of air through the box.
d. Closed cell polymer insulation may be used instead of the fiberglass described above if UL 25/50 labeled.
e. The casing shall be insulated throughout its interior, up to or at least to within 2" of the heating coil connection.
f. Insulation for the heating coil shall enclose the coil casing and tube bends and shall overlap the box internal lining by at least 3".
g. The external insulation shall be as specified in other sections of this specification for duct insulation with full vapor barrier, and shall be field installed unless coil and plenum section is furnished as an integral part of the box.

2. All interior features of the boxes (such as mixing baffles, damper housings, etc.) shall be secured within the casing to avoid excessive movement or rattling with air movement or externally generated vibration. All external features of the terminal units shall be designed not to extend beyond the ends of the unit. (For example, the actuator mounting brackets, etc. shall not extend beyond the plane of the inlet "bulkhead.") The only exception shall be flow sensors installed in the inlet duct connections. Note that if a separate flow station is installed within a frame within the casing, then it shall be so installed not to allow airflow to bypass the flow measurement station.

3. No portion of the VAV terminal shall block access to the flow ring connections on the unit inlets. Note that this will require some manufacturers to mount their control panels further forward than their standard construction.

4. The terminal units shall be constructed with inlet and discharge ductwork connections.

a. The inlet ductwork connections shall extend a minimum of 4 inches from the unit casing including an allowance for the installation of airflow station(s) or probe(s).
b. The discharge connection shall include 1" extension with slip and drive connections for use by the Contractor to secure the discharge ductwork or appurtenances to the unit and shall be reinforced to provide a rigid assembly.

C. CASING LEAKAGE:

1. Assembled units shall be so constructed and sealed to limit air leakage to the following listed quantities at 6" static pressure. The units shall be tested for leakage in both inlets with 6" static pressure imposed on one inlet at a time. The maximum percent leakage from all tests shall be reported.

2. If sealing is required to obtain the leakage performance, seal as for medium pressure ductwork. Hardcast 1602 tape may be used to seal lap joints and flat seams only. Leakage curves or tables will be required as part of the submittal data. The following is the maximum allowable casing leakage, including all components:

Diameter	Maximum CFM (@ 2000 fpm inlet velocity)	Maximum Allowable CFM Casing Leakage
4"-5"-6"	395	8.0
7"-8"	700	14.0
9"-10"	1090	22.0
11"-12"	1570	30.0
13"-14"	2140	40.0

D. ACCESS PLENUM AND DOOR:

1. Single duct units provided with reheat coils also shall be provided with an access section or plenum between the terminal and the coil for coil inspection. The construction of the plenum shall be equal to the quality of materials and workmanship to that of the terminal unit.

2. The access plenum may also be used as a transition, and shall be constructed with a transition angle not to exceed 15 degrees.

3. The access plenum shall contain a minimum of a 12 inch diameter or 12 inch by 12 inch (or full width of unit if less than 12") access door.

   a. The door shall be Ventlok Galvanized Sheet Metal Access Door, Flexmaster Inspector Series, Ward DSP or DSA series, or equal.
   b. The door frame may be bolted, screwed or flanged and sealed to the casing.
   c. The door shall be gasketed and shall be double wall construction or insulated similar to main casing.
   d. The door shall be held in place with latches or other captive retainer devices.
   e. An additional access panel shall be provided immediately downstream of the dampers for inspection and service of the dampers. If the damper assembly is easily removed from the rear of the box, the access size can be reduced to 8" round or 8" x 8" for inspection only.

4. Access doors and frames shall be thermally broken.

E. DAMPER CONSTRUCTION:

1. The damper blades shall be an equivalent of 18-gauge galvanized steel or equal aluminum and shall be securely riveted or bolted through the damper shafts to assure no slippage of the blades.

2. The damper shafts shall be round and operate in sintered bronze self- lubricating bearings. The end of the shaft at the operator end shall be scored in line with the damper blade to reference the damper blade position.

3. Damper shafts penetrating the unit casings shall be sealed against leakage, and bearings shall be installed for protection against wear in the casing penetration.

4. Damper shafts shall be formed of, or cut from solid stock; no hollow shafts will be allowed.

5. The dampers shall seat against gasketed stops or the dampers shall have gasketed edges.

6. Gaskets shall be mechanically fastened to the blades. If the fastening method is not full contact clamping type, then the addition of adhesive to the gasket shall be required.

7. The dampers shall be so constructed to prevent "oil canning" of the damper blade.

8. Leakage curves as a function of pressure shall be supplied as part of the submittal data. The damper actuator linkage, if used, shall be constructed of material of sufficient strength to avoid buckling under extreme loads.

9. Linkages shall not allow play greater than 5 degrees of damper movement. The controls for the dampers shall cause the dampers to fail in the position of last control (freeze in place), or fail to the open position.

F. DAMPER LEAKAGE:

1. The following is the maximum damper leakage allowable for the various size diameter inlets at 6" wg differential pressure. The damper leakage shall not exceed the values listed in the table below at 6" S.P., following ARI 880 Testing Procedures.

Diameter	Maximum CFM (@ 2000 fpm inlet velocity)	Maximum Allowable CFM Damper Leakage
4"-5"-6"	395	6
7"-8"	700	10.5
9"-10"	1090	16.5
11"-12"	1570	20
13"-14"	2140	30

G. UNIT PRESSURE DROP:

1. Single duct unit pressure drop with a 2000 fpm inlet velocity shall be limited to 0.40 inches water gauge unless otherwise specifically noted on terminal box schedule. Units with hot water reheat coils shall have a water side pressure drop of no more than 10 ft of head at the schedule flow.

H. CERTIFICATION:

1. The Unit Manufacturer shall certify that each unit used on this project will perform as specified. Each unit shall bear a tag or decal listing the following specified information:

   a. Test Pressure
   b. Leakage CFM (damper)
   c. Leakage CFM (casing)
   d. Date of Mfg.
   e. Room or area served
   f. Unit size - 6”, 8”, etc.
   g. Calibrated CFM, i.e. 800 CFM

I. MIXING:

1. Terminal units as specified herein shall provide mixing within the units, and not rely upon the discharge ductwork to provide for the completion of the mixing process.

2. The horizontal average temperature of the air as it leaves the terminal unit shall not vary more than 1°F for each 20°F of temperature difference between the two inlet air supplies.

(For example, if the cold supply air is 55°F. and the hot supply air is 95°F., the difference is 40 degrees. The allowable temperature variation of the discharge air is, thus, 2°F.)

3. The temperature of the discharge air shall be measured using a pattern of four vertical, evenly spaced columns, and three horizontal, evenly spaced rows. The rows and columns shall be spaced so that the resulting 12 points shall be at the centers of equal areas. The plane of the points shall be perpendicular to the direction of airflow, within 4 inches of the discharge of the terminal unit, within the discharge ductwork. The three readings in each column shall be averaged to determine compliance with the 1° criteria.

J. FLOW MEASUREMENT:

1. Airflow measurement systems shall measure airflow through the inlet connections within 5% of actual airflow between 2000 fpm and 250 fpm with three straight duct diameters on each air inlet connection.

2. Provide external differential pressure taps separate from the control pressure taps for airflow measurement with a 0”-1”w.g. range.

3. Where flow measurement is indicated to be on the discharge side of the air terminal, the manufacturer is responsible for locating the flow measurement device so that it meets the accuracy requirements above from the minimum to the maximum settings for the box and each deck connection.

4. As a substitute, the manufacturer and contractor may elect to provide an external flow measure device and required duct modifications at no additional cost to the owner.

K. SOUND: Note that the maximum sound levels listed in this paragraph refer to raw sound levels, with no credits taken for the construction.

1. DISCHARGE SOUND

a. Maximum discharge Sound Power Levels at 2000 fpm primary air inlet velocity with 1.5 inch wg inlet static pressure shall not exceed that listed in the following table. No credit for lined discharge duct, branching, flow division, end reflection, room absorption or any other effects shall be allowed.

Octave Band	**Center Frequency (Hz)**	**Sound Power Level (dB re 10-12 Watts)**
2	125	76
3	250	66
4	500	63
5	1000	58
6	2000	60
7	4000	55

2. RADIATED SOUND

a. Maximum discharge Sound Power Levels at 2000 fpm primary air inlet velocity with 1.5 inch wg inlet static pressure shall not exceed that listed in the following table. No credit for ceiling plenum, ceiling tiles, room absorption, or any other effects shall be allowed.

Octave Band	**Center Frequency (Hz)**	**Sound Power Level (dB re 10-12 Watts)**
2	125	72

3	250	67
4	500	64
5	1000	54
6	2000	47
7	4000	45

b. All sound power levels shall be obtained from testing in accordance with ARI-ADC Standard 880 and shall be certified at ARI-880 certification points.

L. HOT WATER COILS:

1. Hot water coils installed in conjunction with single duct terminal units shall be factory installed.

2. Coils shall be one or two row with a maximum of 10 aluminum fins per inch.

3. Air side pressure drop shall be limited to 0.25" wg at box rated cold airflow.

4. Full fin collars shall be provided for accurate fin spacing and maximum fin-to-tube contact.

5. Tubes shall be 1/2 inch diameter seamless copper with a minimum wall thickness of 0.016 inch, tested at 300 psig air pressure under water with a minimum rated burst pressure of 1500 psig.

6. Male sweat-type water connections shall be provided.

7. Side and end plates shall be a minimum of 18-gauge galvanized sheet metal construction.

8. All coils shall be constructed and tested in accordance with UL and/or ARI Standards. The tube ends shall be protected with tube end caps of sheet metal similar to the casing material, and shall be insulated within the caps.

M. CONTROLS:

1. Label the terminal unit and associated room sensors (Temperature, Relative Humidity, CO2, etc) in the field to match the labels in the BAS, or vice versa.

2. Provide unit with single point electrical connection.

3. If DDC controls of another Manufacturer (NOT the terminal unit Manufacturer) are provided for this project, the terminal unit Manufacturer shall be responsible only for the construction of the terminal unit and the installation of internal control components installed at the Manufacturer’s factory, and shall not be responsible for the installation of controls not installed at the terminal unit Manufacturer’s factory, nor shall the Manufacturer be responsible for the performance of the DDC controls.

4. If used, Controls Contractor furnished flow stations shall be furnished, mounted and adjusted by the Controls Contractor with assistance from the Terminal Unit Manufacturer to assure their proper placement.

5. Electronic motors and controllers shall be installed by the terminal unit Manufacturer unless specifically prohibited by the by the controls Manufacturer. In such an event, the controls Manufacturer shall be responsible for the installation of the controls.

6. The controls Manufacturer shall be responsible for the operational performance of the entire system. The terminal unit Manufacturer shall remain responsible only for the performance of the mechanical components of the unit.

7. Refer to the controls drawings and specifications for sequences of operations for the fan powered terminals.

8. Devices using mechanical CFM limiters will not be accepted, nor shall it be necessary to change control components to make airflow rate changes.

N. CONTROLS PERFORMANCE:

1. Units shall be capable of controlling air volume to within plus or minus 5% of air volume setpoint, as determined by the zone temperature sensor demand with variations in inlet pressures from 0.10” to 6” w.g.

2. Assemblies shall be able to be reset to any airflow between zero and the maximum cfm shown on Drawings.

3. To allow for maximum flexibility and future changes, it shall be necessary to make only simple screwdriver or keyboard adjustments to arrange each unit for any maximum airflow within the ranges for each inlet size as scheduled on the Drawings.

4. The control devices shall be designed to maintain the desired flow regardless of inlet flow deflection.

5. Tristate type terminal unit actuators may be used, but they must be automatically, regularly reset/recalibrated so that the actuator position indicated in the BAS is correct.

2.3 VARIABLE OR CONSTANT VOLUME DUAL DUCT TERMINAL UNITS:

A. The Contractor shall provide pressure independent dual duct variable air volume control assemblies as scheduled with integral attenuator/mixers of the sizes, capacities, and configurations shown on the drawings.

B. CASING CONSTRUCTION:

1. The units shall be constructed of a minimum of 22-gauge galvanized steel and internally lined with a minimum of 1 inch thick, three pound per cubic foot density insulation.

   a. The insulation shall be foil faced with the edges and seams sealed or "captured", encapsulating all fibers of the insulation.
   b. Insulation shall meet NFPA-90A requirements for flame spread and smoke generation and UL-181 requirements for anti-erosion, corrosion and fungus properties.
   c. The insulation shall be neatly installed with no rough edges to interrupt the smooth flow of air through the box.
   d. Closed cell polymer insulation may be used instead of the fiberglass described above if UL 25/50 labeled.
   e. The casing shall be insulated throughout its interior.
   f. The external insulation shall be as specified in other sections of this specification for duct insulation with full vapor barrier, and shall be field installed unless coil and plenum section is furnished as an integral part of the box.

2. All interior features of the boxes (such as mixing baffles, damper housings, etc.) shall be secured within the casing to avoid excessive movement or rattling with air movement or externally generated vibration. All external features of the terminal units shall be designed not to extend beyond the ends of the unit. (For example, the actuator mounting brackets, etc. shall not extend beyond the plane of the inlet "bulkhead.") The only exception shall be flow sensors installed in the inlet duct connections. Note that if a separate flow station is installed within a frame within the casing, then it shall be so installed not to allow airflow to bypass the flow measurement station.

3. No portion of the VAV terminal shall block access to the flow ring connections on the unit inlets. Note that this will require some manufacturers to mount their control panels further forward than their standard construction.

4. The terminal units shall be constructed with inlet and discharge ductwork connections.

   a. The inlet ductwork connections shall extend a minimum of 4 inches from the unit casing including an allowance for the installation of airflow station(s) or probe(s).
   b. The discharge connection shall include 1" extension with slip and drive connections for use by the Contractor to secure the discharge ductwork or appurtenances to the unit and shall be reinforced to provide a rigid assembly.

C. CASING LEAKAGE:

1. Assembled units shall be so constructed and sealed to limit air leakage to the following listed quantities at 6" static pressure. The units shall be tested for leakage in both inlets with 6" static pressure imposed on one inlet at a time. The maximum percent leakage from all tests shall be reported.

2. If sealing is required to obtain the leakage performance, seal as for medium pressure ductwork Hardcast 1602 tape may be used to seal lap joints and flat seams only. Leakage curves or tables will be required as part of the submittal data. The following is the maximum allowable casing leakage including all components:

Diameter	**Maximum CFM (@ 2000 fpm inlet velocity)**	**Maximum Allowable CFM Casing Leakage**
4"-5"-6"	395	8.0
7"-8"	700	14.0
9"-10"	1090	22.0
11"-12"	1570	30.0
13"-14"	2140	40.0

D. ACCESS PLENUM AND DOOR:

1. An access panel shall be provided immediately downstream of the dampers for inspection and service of the dampers.

2. The access doors shall be a minimum of a 12 inch diameter or 12 inch by 12 inch (or full width of unit if less than 12").

   a. The door shall be Ventlok Galvanized Sheet Metal Access Door, Flexmaster Inspector Series, Ward DSP or DSA series, or equal.
   b. The door frame may be bolted, screwed or flanged and sealed to the casing.
   c. The door shall be gasketed and shall be double wall construction or insulated similar to main casing.
   d. The door shall be held in place with latches or other captive retainer devices.
   e. If the damper assembly is easily removed from the rear of the box, the access size can be reduced to 8" round or 8" x 8" for inspection only.

3. Access doors and frames shall be thermally broken.

E. DAMPER CONSTRUCTION:

1. The damper blades shall be an equivalent of 18-gauge galvanized steel or equal aluminum and shall be securely riveted or bolted through the damper shafts to assure no slippage of the blades.

2. The damper shafts shall be round and operate in sintered bronze self- lubricating bearings. The end of the shaft at the operator end shall be scored in line with the damper blade to reference the damper blade position.

3. Damper shafts penetrating the unit casings shall be sealed against leakage, and bearings shall be installed for protection against wear in the casing penetration.

4. Damper shafts shall be formed of, or cut from solid stock; no hollow shafts will be allowed.

5. The dampers shall seat against gasketed stops or the dampers shall have gasketed edges. Gaskets shall be mechanically fastened to the blades. If the fastening method is not full contact clamping type, then the addition of adhesive to the gasket shall be required.

6. The dampers shall be so constructed to prevent "oil canning" of the damper blade.

7. The units shall be tested for leakage in both inlets with 6" static pressure imposed on one inlet at a time. The maximum percent leakage from all tests shall be reported. Leakage curves as a function of pressure shall be supplied as part of the submittal data. The damper actuator linkage, if used, shall be constructed of material of sufficient strength to avoid buckling under extreme loads. Also, linkages shall not allow play greater than 5 degrees of damper movement. The controls for the dampers shall cause the dampers to fail in the position of last control (freeze in place), or fail to the open position.

F. DAMPER LEAKAGE:

1. The following is the maximum damper leakage allowable for the various size diameter inlets at 6" wg differential pressure. The damper leakage shall not exceed the values listed in the table below at 6" S.P., following ARI 880 Testing Procedures.

Diameter	**Maximum CFM (@ 2000 fpm inlet velocity)**	**Maximum Allowable CFM Damper Leakage**
4"-5"-6"	395	6
7"-8"	700	10.5
9"-10"	1090	16.5
11"-12"	1570	20
13"-14"	2140	30

G. UNIT PRESSURE DROP:

1. For dual duct units with an integral attenuator-mixer, but with no other accessories, the static pressure across the assembly with an equivalent 2000 fpm inlet velocity through one inlet shall not exceed 0.5 inches water gauge, with the total flow through either inlet.

H. CERTIFICATION:

1. The Unit Manufacturer shall certify that each unit used on this project will perform as specified. Each unit shall bear a tag or decal listing the following specified information:

   a. Test Pressure
   b. Leakage CFM (damper)

c. Leakage CFM (casing)
d. Date of Mfg.
e. Room or area served
f. Unit size - 6”, 8”, etc.
g. Calibrated CFM, i.e. 800 CFM

I. MIXING:

1. Terminal units as specified herein shall provide mixing within the units, and not rely upon the discharge ductwork to provide for the completion of the mixing process.

2. The horizontal average temperature of the air as it leaves the terminal unit shall not vary more than 1°F for each 20°F. of temperature difference between the two inlet air supplies. (For example, if the cold supply air is 55°F. and the hot supply air is 95°F., the difference is 40 degrees. The allowable temperature variation of the discharge air is, thus, 2°F.)

3. The temperature of the discharge air shall be measured using a pattern of four vertical, evenly spaced columns, and three horizontal, evenly spaced rows. The rows and columns shall be spaced so that the resulting 12 points shall be at the centers of equal areas. The plane of the points shall be perpendicular to the direction of airflow, within 4 inches of the discharge of the terminal unit, within the discharge ductwork. The three readings in each column shall be averaged to determine compliance with the 1° criteria.

J. FLOW MEASUREMENT:

1. Airflow measurement systems shall measure airflow through the inlet connections with 5% of actual airflow between 2000 fpm and 250 fpm with three straight duct diameters on each air inlet connection.

2. Provide external differential pressure taps separate from the control pressure taps for airflow measurement with a 0”-1”w.g. range.

3. Where flow measurement is indicated to be on the discharge side of the air terminal, the manufacturer is responsible for locating the flow measurement device so that it meets the accuracy requirements above from the minimum to the maximum settings for the box and each deck connection.

4. As a substitute, the manufacturer and contractor may elect to provide an external flow measure device and required duct modifications at no additional cost to the owner.

K. SOUND: Note that the maximum sound levels listed in this paragraph refer to raw sound levels, with no credits taken for the construction.

1. DISCHARGE SOUND

a. Maximum discharge Sound Power Levels at 2000 fpm primary air inlet velocity with 1.5 inch wg inlet static pressure shall not exceed that listed in the following table. No credit for lined discharge duct, branching, flow division, end reflection, room absorption or any other effects shall be allowed.

Octave Band	**Center Frequency (Hz)**	**Sound Power Level (dB re 10-12 Watts)**
2	125	76
3	250	66
4	500	63
5	1000	58

6	2000	60
7	4000	55

2. RADIATED SOUND

    a. Maximum discharge Sound Power Levels at 2000 fpm primary air inlet velocity with 1.5 inch wg inlet static pressure shall not exceed that listed in the following table. No credit for ceiling plenum, ceiling tiles, room absorption, or any other effects shall be allowed.

Octave Band	Center Frequency (Hz)	Sound Power Level (dB re 10-12 Watts)
2	125	72
3	250	67
4	500	64
5	1000	54
6	2000	47
7	4000	45

    b. All sound power levels shall be obtained from testing in accordance with ARI-ADC Standard 880 and shall be certified at ARI-880 certification points.

L. CONTROLS:

1. Provide unit with single point electrical connection.

2. If DDC controls of another Manufacturer (NOT the terminal unit Manufacturer) are provided for this project, the terminal unit Manufacturer shall be responsible only for the construction of the terminal unit and the installation of internal control components installed at the Manufacturer's factory, and shall not be responsible for the installation of controls not installed at the terminal unit Manufacturer's factory, nor shall the Manufacturer be responsible for the performance of the DDC controls.

3. If used, Controls Contractor furnished flow stations shall be furnished, mounted and adjusted by the Controls Contractor with assistance from the Terminal Unit Manufacturer to assure their proper placement.

4. Electronic motors and controllers shall be installed by the terminal unit Manufacturer unless specifically prohibited by the by the controls Manufacturer. In such an event, the controls Manufacturer shall be responsible for the installation of the controls.

5. The controls Manufacturer shall be responsible for the operational performance of the entire system. The terminal unit Manufacturer shall remain responsible only for the performance of the mechanical components of the unit.

6. Refer to the controls drawings and specifications for sequences of operations for the fan powered terminals.

7. Devices using mechanical CFM limiters will not be accepted, nor shall it be necessary to change control components to make airflow rate changes.

M. CONTROLS PERFORMANCE:

1. Units shall be capable of controlling air volume to within plus or minus 5% of air volume setpoint, as determined by the zone temperature sensor demand with variations in inlet pressures from 0.10" to 6" w.g.

2. Assemblies shall be able to be reset to any airflow between zero and the maximum cfm shown on Drawings.

3. To allow for maximum flexibility and future changes, it shall be necessary to make only simple screwdriver or keyboard adjustments to arrange each unit for any maximum airflow within the ranges for each inlet size as scheduled on the Drawings.

4. The control devices shall be designed to maintain the desired flow regardless of inlet flow deflection.

2.4 FAN-POWERED VARIABLE OR CONSTANT VOLUME TERMINAL UNITS:

A. Units shall be constant air volume secondary, variable air volume primary air distribution assemblies complete with casing, insulation, fan, hot water heating coil, dampers, actuators, controls, transformers, and other appurtenances required for a complete installation. Units shall be of the sizes, configurations, and capacities scheduled.

B. CASING CONSTRUCTION:

1. Casing assembly shall be 20-gauge (minimum) galvanized steel.

2. Interior surfaces of casing shall be acoustically and thermally lined with a 1” thick, 1 1/2-pound density glass fiber insulation with a smooth black coated mat finish equal to Manville “Linacoustic” Fiber Glass Duct Liner or IMCOA closed cell polymer material of equal thermal value.

3. Insulation shall meet NFPA-90A requirements for flame spread and smoke generation and UL-181 requirements for anti-erosion, corrosion and fungus properties.

4. The insulation shall be neatly installed with no rough edges to interrupt the smooth flow of air through the box.

5. The casing shall be insulated throughout its interior, up to or at least to within 2" of the heating coil connection.

6. Insulation for the heating coil shall enclose the coil casing and tube bends and shall overlap the box internal lining by at least 3".

7. The external insulation shall be as specified in other sections of this specification for duct insulation with full vapor barrier, and shall be field installed unless coil and plenum section is furnished as an integral part of the box.

8. All interior features of the boxes (such as mixing baffles, damper housings, etc.) shall be secured within the casing to avoid excessive movement or rattling with air movement or externally generated vibration.

9. All external features of the terminal units shall be designed not to extend beyond the ends of the unit. (For example, the actuator mounting brackets, etc. shall not extend beyond the plane of the inlet "bulkhead.") The only exception shall be flow sensors installed in the inlet duct connections. Note that if a separate flow station is installed within a frame within the casing, then it shall be so installed not to allow airflow to bypass the flow measurement station.

10. No portion of the VAV terminal shall block access to the flow ring connections on the unit inlets. Note that this will require some manufacturers to mount their control panels further forward than their standard construction.

11. The terminal units shall be constructed with inlet and discharge ductwork connections.

a. The inlet ductwork connections shall extend a minimum of 4 inches from the unit casing including an allowance for the installation of airflow station(s) or probe(s).
b. The discharge connection shall include 1" extension with slip and drive connections for use by the Contractor to secure the discharge ductwork or appurtenances to the unit and shall be reinforced to provide a rigid assembly.

C. CASING LEAKAGE:

1. Assembled units shall be so constructed and sealed to limit air leakage to the following listed quantities at 6" static pressure. The units shall be tested for leakage in both inlets with 6" static pressure imposed on one inlet at a time. The maximum percent leakage from all tests shall be reported.

2. If sealing is required to obtain the leakage performance, seal as for medium pressure ductwork. Hardcast 1602 tape may be used to seal lap joints and flat seams only. Leakage curves or tables will be required as part of the submittal data.

3. Casing leakage shall not exceed 2% of the box's maximum scheduled CFM at 6" static pressure.

D. ACCESS PLENUM AND DOOR:

1. Single duct units provided with reheat coils also shall be provided with an access section or plenum between the terminal and the coil for coil inspection. The construction of the plenum shall be equal to the quality of materials and workmanship to that of the terminal unit.

2. The access plenum may also be used as a transition, and shall be constructed with a transition angle not to exceed 15 degrees.

3. The access plenum shall contain a minimum of a 12 inch diameter or 12 inch by 12 inch (or full width of unit if less than 12") access door.

   a. The door shall be Ventlok Galvanized Sheet Metal Access Door, Flexmaster Inspector Series, Ward DSP or DSA series, or equal.
   b. The door frame may be bolted, screwed or flanged and sealed to the casing.
   c. The door shall be gasketed and shall be double wall construction or insulated similar to main casing.
   d. The door shall be held in place with latches or other captive retainer devices.
   e. An additional access panel shall be provided immediately downstream of the dampers for inspection and service of the dampers. If the damper assembly is easily removed from the rear of the box, the access size can be reduced to 8" round or 8" x 8" for inspection only.

4. Access doors and frames shall be thermally broken.

E. DAMPER CONSTRUCTION:

1. The damper blades shall be an equivalent of 18-gauge galvanized steel or equal aluminum and shall be securely riveted or bolted through the damper shafts to assure no slippage of the blades.

2. The damper shafts shall be round and operate in Sintered Bronze self- lubricating bearings. The end of the shaft at the operator end shall be scored in line with the damper blade to reference the damper blade position.

3. Damper shafts penetrating the unit casings shall be sealed against leakage, and bearings shall be installed for protection against wear in the casing penetration.

4. Damper shafts shall be formed of, or cut from solid stock; no hollow shafts will be allowed.

5. The dampers shall seat against gasketed stops or the dampers shall have gasketed edges.

6. Gaskets shall be mechanically fastened to the blades. If the fastening method is not full contact clamping type, then the addition of adhesive to the gasket shall be required.

7. The dampers shall be so constructed to prevent "oil canning" of the damper blade.

8. Leakage curves as a function of pressure shall be supplied as part of the submittal data. The damper actuator linkage, if used, shall be constructed of material of sufficient strength to avoid buckling under extreme loads.

9. Linkages shall not allow play greater than 5 degrees of damper movement. The controls for the dampers shall cause the dampers to fail in the position of last control (freeze in place), or fail to the open position.

F. DAMPER LEAKAGE:

1. The following is the maximum damper leakage allowable for the various size diameter inlets at 6" wg differential pressure. The damper leakage shall not exceed the values listed in the table below at 6" S.P., following ARI 880 Testing Procedures.

Diameter	Maximum CFM (@ 2000 fpm inlet velocity)	Maximum Allowable CFM Damper Leakage
4"-5"-6"	395	6
7"-8"	700	10.5
9"-10"	1090	16.5
11"-12"	1570	20
13"-14"	2140	30

G. FANS:

1. The fan shall be forward curved, centrifugal, direct-drive type with 120-volt/1-phase motor and SCR controller for air flow adjustment from 60 - 100% of nominal box scheduled CFM.

2. The fan and motor assembly shall be internally suspended and isolated from the casing on rubber in shear isolators.

3. The fan and motor assembly shall be easily accessible through access panels without disassembling the entire unit.

4. The SCR controller and fan motor shall be harmonically balanced to reduce electrical noise. Fan assembly shall include an anti-backward rotation device.

H. FLOW MEASUREMENT:

1. Airflow measurement systems shall measure airflow through the inlet connections with 5% of actual airflow between 2000 fpm and 250 fpm with three straight duct diameters on each air inlet connection.

2. Provide external differential pressure taps separate from the control pressure taps for airflow measurement with a 0”-1”w.g. range.

3. Where flow measurement is indicated to be on the discharge side of the air terminal, the manufacturer is responsible for locating the flow measurement device so that it meets the accuracy requirements above from the minimum to the maximum settings for the box and each deck connection.

4. As a substitute, the manufacturer and contractor may elect to provide an external flow measure device and required duct modifications at no additional cost to the owner.

I. HOT WATER COILS:

1. Hot water coils installed in conjunction with single duct terminal units shall be factory installed.

2. Coils shall be one or two row with a maximum of 10 aluminum fins per inch.

3. Air side pressure drop shall be limited to 0.25" wg at box rated cold airflow.

4. Full fin collars shall be provided for accurate fin spacing and maximum fin-to-tube contact.

5. Tubes shall be 1/2 inch diameter seamless copper with a minimum wall thickness of 0.016 inch, tested at 300 psig air pressure under water with a minimum rated burst pressure of 1500 psig.

6. Male sweat-type water connections shall be provided.

7. Side and end plates shall be a minimum of 18-gauge galvanized sheet metal construction.

J. CONTROLS:

1. Provide unit with single point electrical connection.

2. If DDC controls of another Manufacturer (NOT the terminal unit Manufacturer) are provided for this project, the terminal unit Manufacturer shall be responsible only for the construction of the terminal unit and the installation of internal control components installed at the Manufacturer’s factory, and shall not be responsible for the installation of controls not installed at the terminal unit Manufacturer’s factory, nor shall the Manufacturer be responsible for the performance of the DDC controls.

3. If used, Controls Contractor furnished flow stations shall be furnished, mounted and adjusted by the Controls Contractor with assistance from the Terminal Unit Manufacturer to assure their proper placement.

4. Electronic motors and controllers shall be installed by the terminal unit Manufacturer unless specifically prohibited by the by the controls Manufacturer. In such an event, the controls Manufacturer shall be responsible for the installation of the controls.

5. The controls Manufacturer shall be responsible for the operational performance of the entire system. The terminal unit Manufacturer shall remain responsible only for the performance of the mechanical components of the unit.

6. Refer to the controls drawings and specifications for sequences of operations for the fan powered terminals.

7. Devices using mechanical CFM limiters will not be accepted, nor shall it be necessary to change control components to make airflow rate changes.

K. CONTROLS PERFORMANCE:

1. Units shall be capable of controlling air volume to within plus or minus 5% of air volume setpoint, as determined by the zone temperature sensor demand with variations in inlet pressures from 0.10” to 6” w.g.

2. Assemblies shall be able to be reset to any airflow between zero and the maximum cfm shown on Drawings.

3. To allow for maximum flexibility and future changes, it shall be necessary to make only simple screwdriver or keyboard adjustments to arrange each unit for any maximum airflow within the ranges for each inlet size as scheduled on the Drawings.

4. The control devices shall be designed to maintain the desired flow regardless of inlet flow deflection.

PART 3 - EXECUTION

3.1 TESTING PRIOR TO INSTALLATION:

A. Refer to Submittal Paragraph (in Part 1 of this section) for sample unit testing prior to shipment.

B. SHIPMENT TESTING:

1. A minimum of ten percent (10%) of each size of the terminal units (but no less than one unit of each size used) may be tested for conformance to this specification, at the Owner's discretion. The Contractor shall allow sufficient time during construction and space for the Owner’s TAB Consultant to perform all testing as may be required.

C. UNIT NON-PERFORMANCE:

1. If the results of the SHIPMENT TESTING show that any of the units do not perform as specified, then an additional ten percent (10%) of each size (but no less than one unit of a size, unless 100% of the size has been tested) of the units shall be tested. If this testing, in the Owner’s opinion, shows that ten percent (10%) or more of the units tested do not perform as specified, then one hundred percent (100%) of all sizes of the units shall be tested for conformance with these specifications. The results of that testing shall be reviewed carefully between the Contractor, Manufacturer, the Owner’s construction administrator(s), and the Owner’s design engineer(s). A method of repair or replacing the units will be negotiated. The Owner, however, shall maintain the right of final approval of any proposed solution.

2. Should for any reason the testing described above under "SUBMITTAL" and "SHIPMENT TESTING" prove that any of the units do not perform as specified, the Contractor shall be responsible for all subsequent labor, travel, travel expenses, and incidental expenses, penalties, or other costs attendant to any additional testing as described under "UNIT NON-PERFORMANCE", or as required to prove that the units perform as specified. This shall include, but not be limited to, the labor, travel and reasonable incidental expenses of not only the Contractor and Owner’s TAB Consultant, but also those incurred by the Owner as may be specifically required for this purpose. The expenses to be reimbursed to the Owner shall be labor at a rate of $300 per day or any portion of a day, plus travel and travel expenses at actual cost, plus reasonable incidental expenses at actual cost.

3. All coils shall be constructed and tested in accordance with UL and/or ARI Standards. The tube ends shall be protected with tube end caps of sheet metal similar to the casing material, and shall be insulated within the caps.

3.2 INSTALLATION

A. Refer also to requirements included in Part 2 of this specification.

B. Install in accordance with Manufacturer's instructions.

C. Provide ceiling access doors or locate units above easily removable ceiling components.

D. Support units individually from structure. Do not support from adjacent ductwork.

E. Connect to ductwork in accordance with Division 23 Ductwork specification.

F. All terminal units shall be installed with manufacturer's minimum straight duct inlet requirements or 3 duct diameters of straight run, whichever is larger.

G. Maintain NEC and manufacturer's recommended clearances for control enclosures.

END OF SECTION 23 36 00